

Experiments on turbulent natural convection in an enclosed tall cavity

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Abstract

Experiments have been undertaken to investigate the natural convection of air in a tall differentially heated rectangular cavity (2.18 m high by 0.076 m wide by 0.52 m in depth). They were performed with temperature differentials between the vertical plates of 19.6°C and 39.9°C, giving Rayleigh numbers based on the width of 0.86×10^6 and 1.43×10^6 . Under these conditions the flow in the core of the cavity is fully turbulent and property variations with temperature are comparatively small. A previously used experimental rig has been modified, by fitting partially conducting top and bottom walls and outer guard channels, to provide boundary conditions which avoid the inadequately defined sharp changes in temperature gradient and other problems associated with insufficient insulation on nominally adiabatic walls. Mean and turbulent temperature and velocity variations within the cavity have been measured, together with heat fluxes and turbulent shear stresses. The temperature and flow fields were found to be closely two-dimensional, except close to the front and back walls, and anti-symmetric across the diagonal of the cavity. The partially conducting roof and floor provide locally unstable thermal stratification in the wall jet flows there, which enhances the turbulence as the flow moves towards the temperature controlled plates. The results provide a greatly improved benchmark for the testing of turbulence models in this low turbulence Reynolds number flow. © 2000 Elsevier Science Inc. All rights reserved.

Keywords: Natural convection; Turbulence; Heat transfer; Closed cavity

1. Introduction

Natural convection within enclosures occurs in many practical situations ranging from simple space heating of domestic rooms to parts of industrial and nuclear installations. In typical applications, although much of the flow is turbulent, there are extensive regions with a low turbulent Reynolds number, and often significant regions of laminar flow or regions of stagnation caused by local stratification. Moreover, when the heat which drives the flow is introduced or extracted over approximately vertical boundaries, the thermal conditions there are close to constant heat flux with distance from the wall, but the shear stress varies rapidly and changes sign close to the wall. For these reasons, simple turbulence modelling of such flows, and particularly the use of standard wall laws, is totally inappropriate.

Despite the widespread occurrence of natural convection in enclosed regions, the availability of good quality experimental data involving both temperature and velocity fields is still very limited, e.g., Cheesewright et al. (1986), Dafa'Alla and Betts

(1996), Mergui et al. (1993), and Tian (1997), although the last is for a very low level of turbulent fluctuations. Data on temperature fields and heat flux is extensive, ranging from the well-known experiments of MacGregor and Emery (1969) to the large models of domestic rooms investigated by Yguel (1988). However, the low velocities involved are difficult to measure, as hot wire probes generate their own local convection and are insensitive to the flow reversals present in the flow. Laser Doppler anemometry (LDA), with Bragg shift to detect flow reversals, or an equivalent technique, is therefore essential for the measurement of velocity fields. Variations in refractive index of liquids with temperature cause difficulties with bending of the laser beams and, in a turbulent flow, a consequent movement of the beams and frequent loss of the measuring volume. Velocity measurements are therefore most conveniently performed in gases, normally air.

The experimental situation being investigated here is that of a simple rectangular box-shaped cavity, with opposing vertical walls held at different temperatures. The existence of turbulence within such a cavity depends on aspect ratio $A = H/W$ as well as Rayleigh number. For the nearly square cavity of Mergui et al. (1993) the flow was only transitional with a stratified quiescent flow in the core, even at a Rayleigh number based on cavity width W of $Ra = 2.3 \times 10^9$. Similarly, turbulence was even less extensive in the square cavity of Tian (1997)

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Notation		Ra	Rayleigh number ($g\beta\Delta TW^3/\nu\alpha$)
A	aspect ratio H/W	T, T_{rms}	mean and rms components of temperature
D	depth of cavity	U, u, V, v	mean and rms horizontal and vertical velocities
H	height of cavity	W	width of cavity
		x, y, z	horizontal, vertical and depth-wise coordinates

at $Ra = 1.6 \times 10^9$. In contrast, in the tall cavity of Dafa'Alla and Betts (1996) with $A = 28.6$ and $Ra = 10^6$, the flow in the core region flow was fully turbulent with turbulence levels over half the maximum mean vertical velocity. This is consistent with the numerical work of Le Quéré and Alziary (1985), which demonstrated the rapid reduction of the Rayleigh number, based on cavity width, required for transition as the aspect ratio is increased. Thus, the use of a tall cavity reduces the width W and/or the temperature difference between the vertical plates needed to ensure turbulent flow within the cavity. The smaller width then means that only moderate values of the third dimension, cavity depth, are required to produce approximately two-dimensional flow in the region either side of the centre-plane of the cavity (cf. Cheesewright et al. (1986) with $A = 5$). Similarly a moderate temperature difference between the vertical walls ensures that problems of property variations within the gas can be neglected.

Most experimental work has attempted to provide data under idealized thermal boundary conditions, such as constant temperature or constant heat flux active walls and adiabatic or perfectly conducting passive ones. For example, Dafa'Alla and Betts (1996) attempted to impose a constant heat flux on the hot wall, with an isothermal cold wall and all other walls insulated. As discussed by Allard (1993) and Fusegi and Hyun (1994), all solid insulators have thermal conductivities greater than air, so that it was inevitably found impossible to reproduce the ideals specified. Dafa'Alla and Betts encountered particular difficulties at the junction between the electrically heated hot plate and the nominally adiabatic top and bottom walls. These problems were typical of all attempts to reproduce idealized boundary conditions, and consequently, when the results were used by Ince et al. (1993) to test low-Reynolds-number turbulence models, measured wall temperatures had to be used as input rather than the ideals specified.

Although the results of Ince et al. (1993) were encouraging to the turbulence modellers in general terms, the uncertainty in boundary layer definition in regions of rapidly changing temperature gradient (e.g., at one end of the nominally adiabatic top and bottom surfaces) meant that one does not know whether the relatively minor differences between experiment and computation should be attributed to the turbulence model or the problems with the experiment. The present experiments were therefore designed to overcome these difficulties and provide an appropriate benchmark for modellers. The apparatus of Dafa'Alla and Betts (1996) has been rebuilt to ensure well-defined, readily measurable, and smoothly varying temperatures within the boundary walls. Conduction within the walls becomes part of the problem, but there is now a clear unambiguous specification. Moreover, the active walls are now both effectively isothermal, the internal surface temperatures of the top and bottom vary smoothly, heat loss through the front and back walls is small, and the depth of the cavity is sufficient to produce effectively two-dimensional flow over most of the enclosure volume. The results have been used as a benchmark for testing the numerical predictions of turbulence models at the 5th and 7th ERCOFTAC workshops on refined flow modelling (1996, 1998) and are presented here with more detail for a wider audience. A summary of the results at mid-

height of the cavity is given in Table 1, near the end of this paper.

2. Experimental apparatus and instrumentation

2.1. Apparatus

The apparatus is a modification to that of Dafa'Alla and Betts (1996), and is described in detail in Bokhari (1996). The internal dimensions of the cavity shown diagrammatically on Fig. 1(a) are 2.18 m high (H), 0.076 m wide between the hot and cold walls (W) and 0.52 m deep (D). The hot and cold plates are of polished aluminium backed by temperature controlled water jackets and the water passes upwards along channels behind each plate, with independent flow control of each channel to ensure equality of temperature across the plates. Temperatures at entry were controlled to $\pm 0.05^\circ\text{C}$ by PID (REX, C10) controllers. For the cold plate it was found to be most convenient to cool the cold tank continuously and control the temperature by means of a supplementary (1 kW) heater inserted in the well-mixed tank. The whole rectangular cavity assembly is supported from a shaft at the back of the hot plate, which is connected to the separate main support frame. The cavity can therefore be tilted if required in the future.

The active plates are oversized relative to the internal dimensions of the cavity as part of the isothermal control, as shown on Fig. 1(b) and (c). The new hot plate is 2.38 m by 0.53 m and is backed by a 10 mm thick water jacket with four channels. The water jacket is then insulated on its back with 1 cm thick closed-cell thermo-flux insulating material and bolted to a cellular wooden frame with a 5 cm thickness of polystyrene behind it, to further reduce heat losses. The cold plate is a refurbishment of that used previously and is 2.40 m by 0.53 m backed by a 4.8-mm water jacket with three channels. It is supported externally from the hot wall mainly at the top and bottom, but with two bolts on the front and back of the cavity to ensure a uniform width to the cavity. The cold water jacket is backed by 5 cm of polystyrene.

The passive front and back walls are of 9.5 mm thick acrylic sheet. The intention is that temperature gradients normal to these faces should be small (close to adiabatic). However with the existing instrumentation and supports, it was not possible to provide full internal guard cavities of the sort used by Mergui et al. (1993) and Tian (1997) without reducing the depth dimension of the main cavity unacceptably. As an alternative, the acrylic walls were clamped externally by square sectioned stainless steel hollow channels, shown on Fig. 1. Water was circulated through these channels from the same tanks as were used for the hot and cold plates and the whole assembly backed by 5 cm polystyrene slabs. In this way an isothermal boundary condition can be imposed in the plane of the active plates right out to and through the passive walls. Although heat loss through the acrylic walls is only minor, the external boundary conditions are fully defined by means of a strip of seven thermocouples which can be attached to the acrylic sheet with good thermal contact at any elevation.

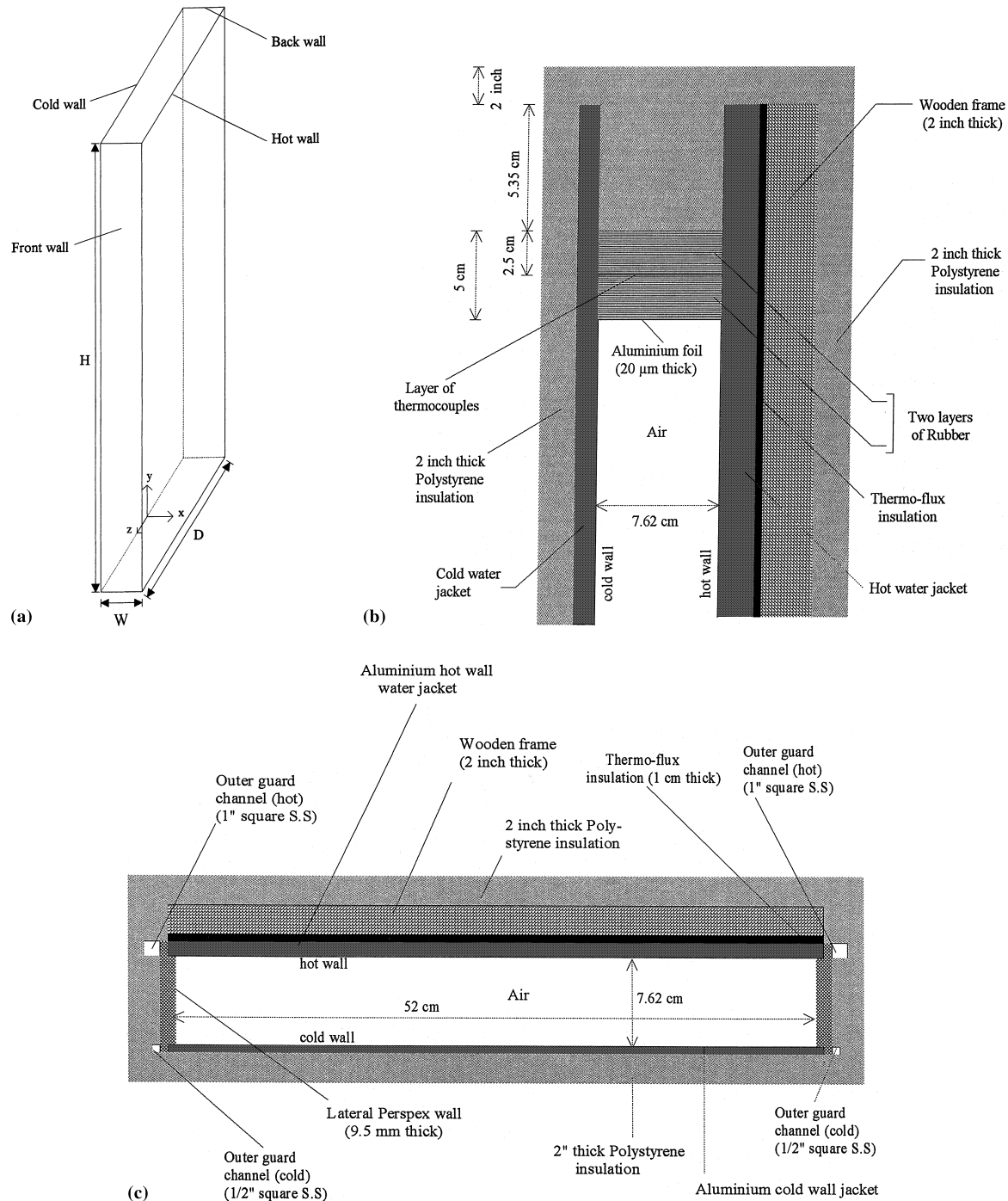


Fig. 1. Diagrams of the experimental apparatus: (a) schematic diagram; (b) vertical section of the top of the cavity; (c) horizontal section of the cavity.

The top and bottom walls are constructed from 5 cm thick natural hard rubber, with a layer of 35 thermocouples imbedded at mid-thickness. They are wedged by the active walls, with good thermal contact ensured by a layer of heat conducting grease. The outer sides of the rubber walls are covered with 10 cm of polystyrene insulation and the inner sides by a thin (10 μm) sheet of adhesive aluminium foil, to reduce radiation absorption. The natural rubber has a thermal conductivity of 0.155 W/mK, which is close to the effective mean conductivity across the cavity under the turbulent conditions

(i.e., Nusselt number times molecular conductivity of air). Consequently, this avoids both the excessive drainage of heat between the active walls, which would occur with “perfectly conducting” walls, and the inadequately defined sharp temperature variations encountered close to an active wall in attempts at adiabatic conditions. As well as providing well defined boundary conditions, the use of the rubber also provides unstable thermally stratified layers adjacent to both the roof and the floor, which oppose the re-laminarization which might otherwise occur there (cf. Cheesewright et al., 1986).

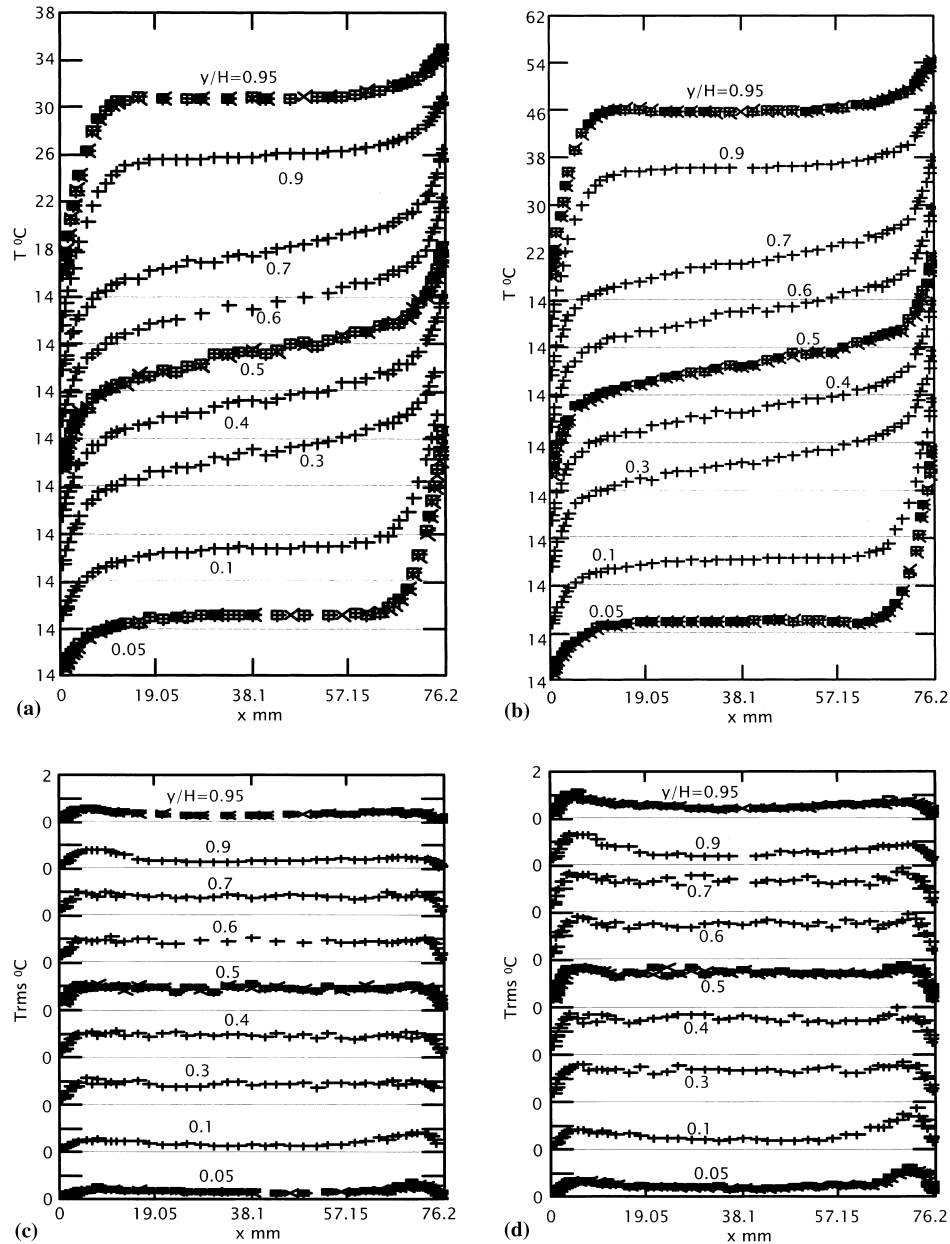


Fig. 2. Temperature traverses across the cavity at various heights: (a) mean temperatures, $Ra = 0.86 \times 10^6$; (b) mean temperatures, $Ra = 1.43 \times 10^6$; (c) rms temperatures, $Ra = 0.86 \times 10^6$; (d) rms temperatures, $Ra = 1.43 \times 10^6$ (+ : $z/W = 0$ mm; $\times$: $z/W = 130$ mm; $\square$: $z/W = 230$ mm).

2.2. Instrumentation

Temperatures in the water circulating in the water jackets and guard channels were monitored by nine thermocouples on the cold side and 21 on the hot. These, like those in the rubber roof and floor and on the outside of the acrylic passive walls, are K-type and connected to switch boxes. A single traversed K-type thermocouple was also used for measurements within the turbulent airflow inside the cavity. The diameter of the Chromel-Alumel wires is $75 \mu\text{m}$ diameter, which is robust enough for long use and has a sufficiently rapid response time, calculated to be 0.07 s (equivalent to 13 Hz). During this work, the response time of the traversing thermocouple was also measured directly by measuring the time history as the bead was intermittently heated by a laser beam. This gave a response time of 0.065 s. Access for the internal thermocouple

traverses is through sealed slots in a vertical acrylic wall and positional accuracy as distance from a wall is estimated to be smaller than the wire thickness.

At each traverse position, 2048 samples were recorded in 128 s. The signal was amplified by a factor of 1000 by a DANA type 2850 DC amplifier before passing to a DISA type 55D26 signal conditioner for final amplification and low pass filtering. The signal was then digitized by a 12 bit OEI ADC unit to provide a digital accuracy of about 0.1°C . This choice of processing conditions reflects the known presence of significant low frequency components in the turbulent flow. During the course of the present work, spectral analyses of both the temperature and velocity records were made. These indicated that the high frequency cut off from the response time and the methods of processing the signals had a negligible effect on the recorded rms values.

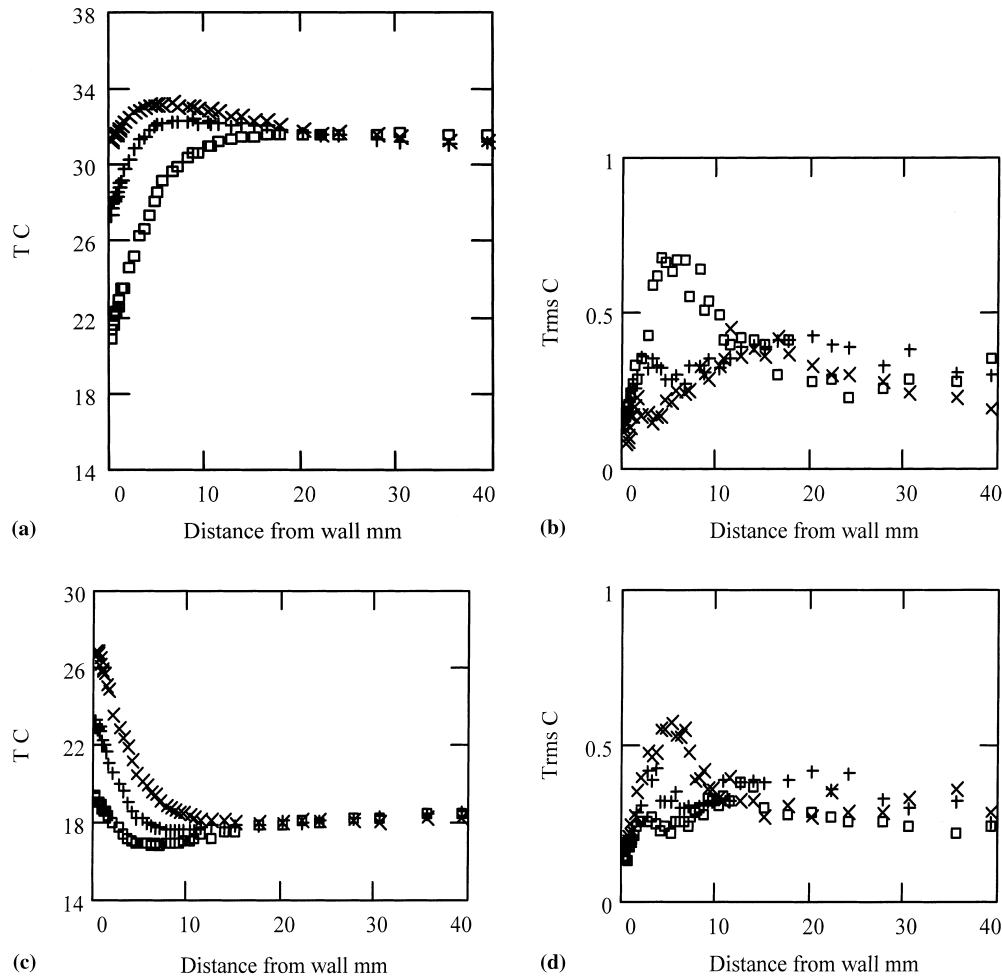


Fig. 3. Centre-plane temperature traverses normal to top and bottom walls at $Ra = 0.86 \times 10^6$: (a) top wall, mean temperatures; (b) top wall, rms temperatures; (c) bottom wall, mean temperatures; (d) bottom wall, rms temperatures ($\square$: $x = 19.1$ mm; $+$: $x = 38.1$ mm; $\times$: $x = 57.2$ mm).

For velocity measurements, a single component LDA system was used with a 15 mW He–Ne laser, in forward scatter mode with dual Bragg shift and a beam expander, which produced a measuring volume diameter of 0.37 mm, 4 mm long. Electrical signals were produced by an avalanche photodiode and processed by an INVENT DSP card. The APD is more sensitive to red light than the standard photodiode, and its use provides a dramatic improvement in the signal-to-noise ratio. Band-pass filtering at 0.1–0.5 MHz with a Bragg shift of 0.1983 MHz (corresponding to a velocity range of -0.46 to 1.38 m/s) was found to be wide enough to cover the full range of instantaneous velocities. The adequacy of these settings was checked by occasionally reversing the Bragg shift, especially around positions of peak mean velocity, and observing that the results were unchanged. Seeding was by smoke from a burning joss stick, which was injected for 2–3 min. After leaving the flow to settle down for about an hour, the smoke is fully diffused with no large particles. Although the particle size is somewhat small (order $1 \mu\text{m}$), which reduces the signal-to-noise ratio, it is adequate for a session of measurements up to 8 h without need of further seeding. Typically 4096 samples were obtained in 5–10 min. For LDA measurements the acrylic wall with sealed slots used with the thermocouple is replaced by an optically plane one. Access to the cavity for the optical measurements is obtained by removing only the minimum amount of insulation required locally. The traversing equipment for

both temperature and velocity measurements was mounted on a Dexion frame which was attached to both the solid floor and the wall. Vibration effects were measured by obtaining a LDA signal from a flaw in a stationary piece of acrylic placed in the convection cell. This indicated that vibration effects were less than 0.1 mm/s.

3. Results and discussion

3.1. Temperatures

Data was obtained for Rayleigh numbers of 0.86×10^6 and 1.43×10^6 based on cavity width, measured air-side wall temperatures and air properties evaluated at mean temperatures. Fig. 2 shows the results of thermocouple traverses across the cavity at nine heights and various front-to-back distances across it. Offset scales are used for each height, the dimensionless values of which is given as a label on each traverse. This data covers vertical planes out to 88% of the cavity depth, and mean temperatures over the different planes at the same height are indistinguishable (Fig. 2(a) and (b)). The same is also true of the rms values (Fig. 2(c) and (d)) although the scatter is inevitably more evident. Traverses normal to the front acrylic wall at mid-height and three equispaced distances across the width showed constant mean temperatures to within

5 mm of the wall and decay of the fluctuating temperature starting about 10 and 15 mm from the wall at the two values of Ra . Similarly vertical traverses on the vertical centreline ($x = 38.1$ mm) normal to the top and bottom boundaries at $z = 0$ and 130 mm were indistinguishable, and the thermocouples imbedded within the rubber indicated the same linear variation of temperature between the plates at all values of z . The temperature results therefore demonstrate an effectively two-dimensional temperature field over the middle 90% of the cavity. Measurements of effects close to the front wall are given in Bokhari (1996). Discussion here will concentrate on the two-dimensional aspects and in particular over the centre-plane $z = 0$ and $Ra = 0.86 \times 10^6$, where fluid property variations are less over the 19°C temperature range.

The first thing one notices about the results on Fig. 2 is the high quality of the anti-symmetry, both across the cavity in the mid-height region ($y/H = 0.5$) and between the top and the bottom ($y/H = 0.05$ and 0.95). A major cause of this was the choice of the rubber top and bottom walls which produced similar thermal stratifications in the wall flows there, which is confirmed by the traverses shown on Fig. 3. These traverses indicate unstably stratified boundary layers with near-wall stratification and thermal fluctuations increasing as the flow travels horizontally. This raised level of fluctuation continues passed the corners into the start region of the vertical flows along the active plates, as can be seen on Fig. 2 at $y/H = 0.95$ on the cold side and $y/H = 0.05$ on the hot side. It can also be seen that, although the variation of temperature at the surface of the top and bottom walls is not exactly linear between the active walls, there are no sudden changes of gradient, which can make interpolation of data so difficult (cf., Dafa'Alla and Betts, 1996). As a consequence, the benchmark boundary conditions can be chosen either as linear on the outer surfaces of the rubber (or at the position of the thermocouples imbedded within it) or by interpolation from the experimental data at the surfaces.

An early examination of the temperature variations normal to the active walls showed that in the first 3 mm from the wall (seven readings) the variation of temperature is closely linear with distance from the wall. Thus, since heat flux is constant there, this region is within the molecular conduction layer and wall temperatures can be accurately obtained by extrapolation, while heat transfer can be determined from the wall gradient. The validity of this way of measuring the surface heat flux had previously been confirmed in the earlier experiments of Ince (1985) and Dafa'Alla and Betts (1996), where the hot wall was heated electrically. From a comparison of these surface temperatures with those obtained from the thermocouples in the water jackets, it became apparent that there was a noticeable thermal resistance on the water-side of the active plates, particularly on the older cold plate. Despite repeated flushing with the anti-corrosion additive Rodol NA1, it was not possible to reduce the temperature difference between the water and the air-side of the cold plate to less than 0.9°C at the lower Rayleigh number, while differences on the hot side were no more than 0.3°C . The temperatures quoted in this paper are therefore the air-side temperatures obtained from extrapolation of the air traverses. These have a maximum variation of $+0.2/-0.4^\circ\text{C}$ about the mean on the cold side (compared with a temperature rise in the flowing water of 0.3°C) and $+/-0.2^\circ\text{C}$ (compared with 0.2°C) on the hot side. This is a comparatively small deviation from isothermal conditions, as is shown on Fig. 4(a). For maximum accuracy of the curve fitting, use was made of an analysis similar to that of Launder and Sharma (1974). This shows that, in a constant heat flux layer with the mean velocity normal to the wall zero, the leading term from the near wall turbulence in a polynomial for mean temperature is the fourth power of distance from the wall. Account can also

be taken of a linear variation of thermal conductivity with temperature by including a calculable second order term (no cube term) in the expansion. One then obtains a fourth order polynomial for mean temperature with no cubic term and only three unknown coefficients. Least squares fitting for this polynomial gave excellent agreement over the first eight data points, with typical standard errors of 0.07°C . Taking into account also uncertainty in positional accuracy and thickness of the thermocouple, the wall temperature gradients from this

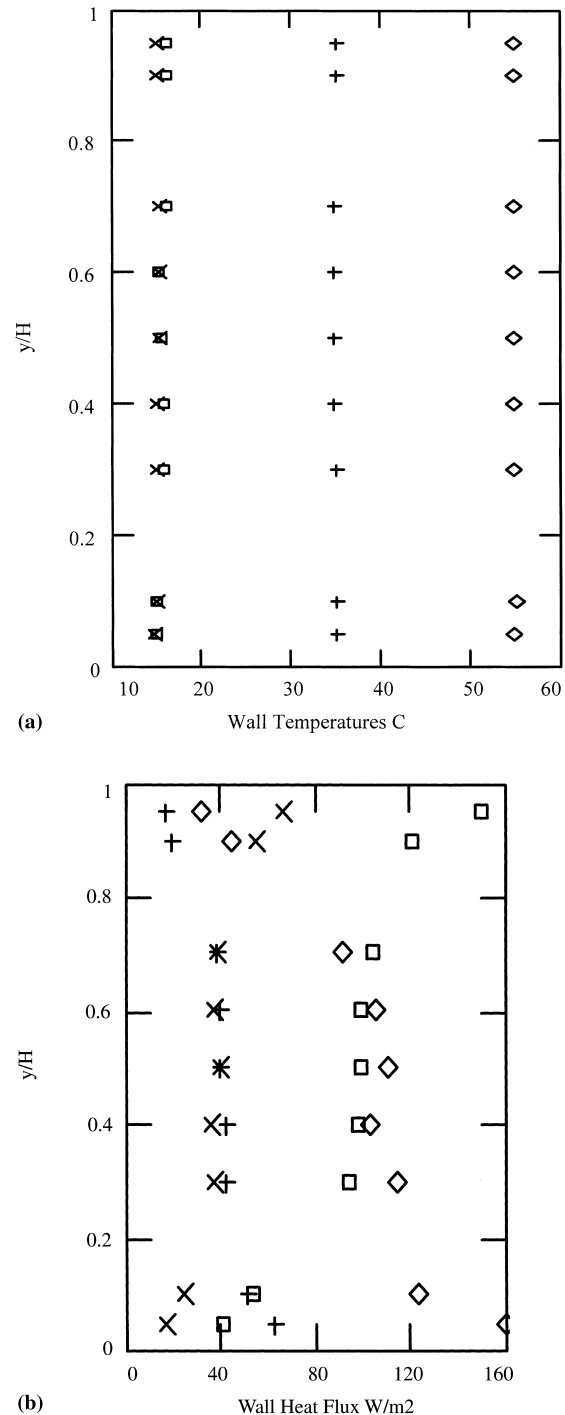


Fig. 4. Variations along vertical centre-line of active walls: (a) mean temperature; (b) normal heat flux: $Ra = 0.86 \times 10^6$ (x: cold wall; +: hot wall), $Ra = 1.43 \times 10^6$ ($\square$ cold wall, $\diamond$ hot wall).

analysis are estimated to be accurate to $\pm 5\%$. Allowing for the difference in thermal conductivity of the air on each side (6% and 12% at the two Rayleigh numbers), the variation of wall heat flux over the height of the active walls can be calculated and is shown on Fig. 4(b). An integration of the fluxes over the height of the cavity indicates heat transfer rates on the hot side 1.1% and 3.3% higher on the hot wall than on the cold. This is within the estimated experimental error and compares very favourably with the 14.7% imbalance in Dafa'Alla and Betts (1996) or 20% of Ziai (1983).

3.2. Velocities

Profiles of mean and fluctuating vertical velocity are presented in Fig. 5. As with temperature, the results are essentially two dimensional, although limitations of optical access meant that this could only be confirmed out to 50% of the cavity depth. Note that the almost linear variation of velocity across

the centre at mid-height is associated with the maximum of fluctuating velocities there. The maximum rms vertical velocity at the centre, where the mean velocity is zero, is about 70% of the maximum mean velocities near the active walls. This is in contrast to the quiescent central region that occurs at lower aspect ratios (e.g., the square cavity of Mergui et al. (1993)) and is associated with interference between the rising and descending boundary layers over the opposite walls. As expected from the temperature fields, anti-symmetry of velocity is good, both across the mid-height traverse and across opposite corners of the cavity. It is also noticeable that the flow undergoes a strong acceleration as it starts to move along the active plates, with peak values reaching maxima at around $y/H = 0.1$ and 0.9 before decelerating to give approximately constant profiles between 0.3 and 0.7 , where the peak velocity is further from the wall. The fluctuating velocities also indicate a local maximum at the start of each active plate, although slightly further out in the cavity.

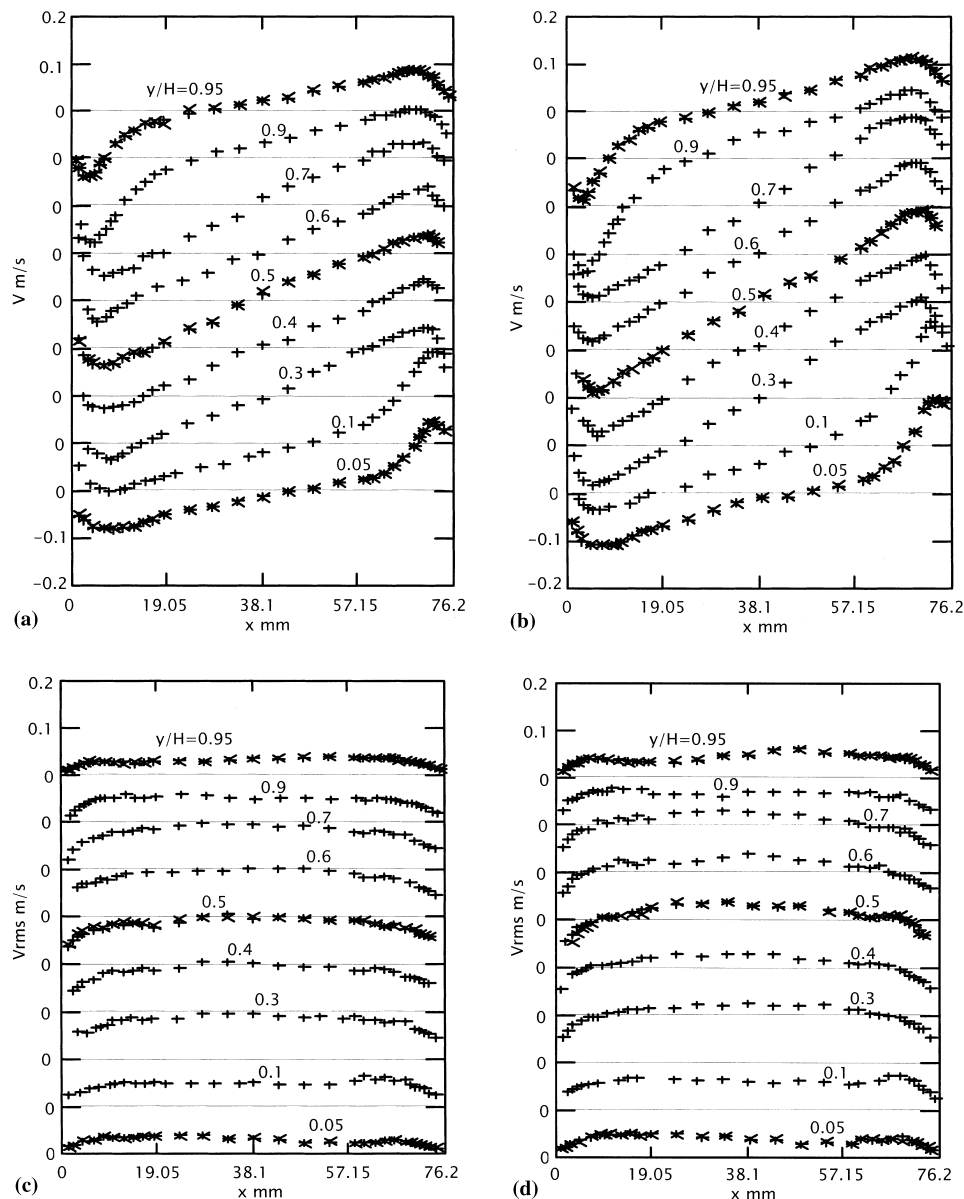


Fig. 5. Traverses of vertical velocity across the cavity at various heights: (a) mean velocities, $Ra = 0.86 \times 10^6$; (b) mean velocities, $Ra = 1.43 \times 10^6$; (c) rms velocities, $Ra = 0.86 \times 10^6$; (d) rms velocities, $Ra = 1.43 \times 10^6$ (+ : $z/W = 0$ mm; $\times$: $z/W = 130$ mm).

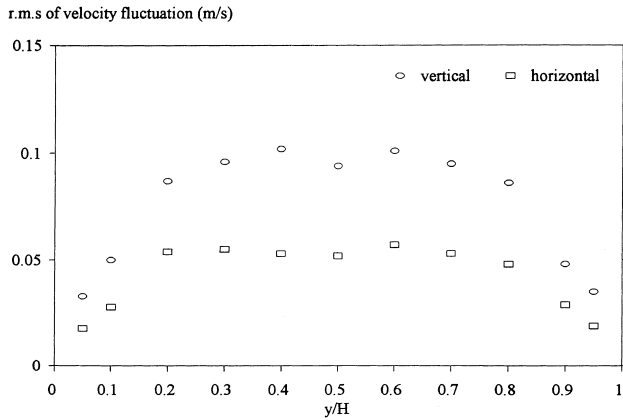


Fig. 6. Vertical and horizontal fluctuating velocities along the vertical centre-line at $Ra = 0.86 \times 10^6$.

Traverses along the vertical centre-line, shown on Fig. 6, reveal that the turbulence level is virtually constant from $y/H = 0.2$ to 0.8 , with vertical velocity fluctuations almost twice the horizontal ones. Turbulence levels drop off towards the top and bottom, but then the horizontal component maintains a constant level of 0.015 m/s (at $Ra = 0.86 \times 10^6$) through the thermally unstable wall-jet type boundary layers close to the top and bottom walls.

Optical access with the current set-up restricts measurements of horizontal velocity components to only half of the cavity width. However, by rotating the single channel laser optics between $+/- 45^\circ$ and differencing the rms values, the distributions of Reynolds stress across that part of the cavity could be obtained and are shown on Fig. 7.

3.3. Derived and other results

Spectra of both temperature and velocity have been obtained from the time records. Both indicated significant roll-off by 1 Hz. Correction of sample temperature data by a numerical amplifying filter to offset the measured response time of 0.07 s only increased the rms temperature value by 4% . Examples of the spectra are given in Bokhari (1996) and digitised

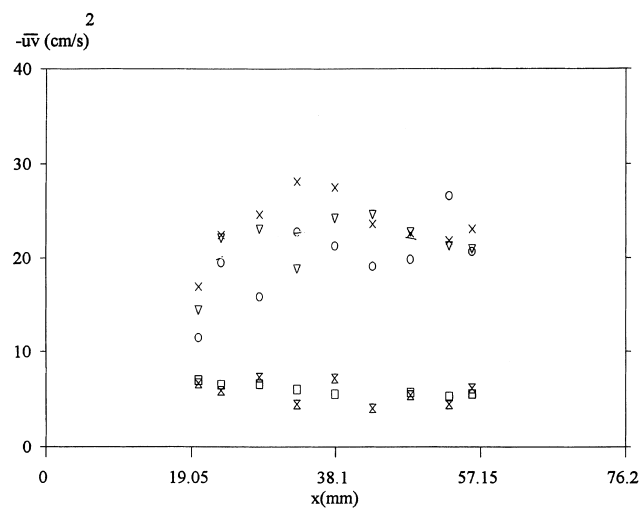


Fig. 7. Variations of Reynolds stress across centre-plane of cavity at various heights, $Ra = 0.86 \times 10^6$ ($\square$, $y/H = 0.1$; $\circ$, $y/H = 0.4$; $\times$, $y/H = 0.5$; ∇ , $y/H = 0.6$; $\times$, $y/H = 0.9$).

samples of time series are available in the ERCOFTAC “Classic” database collection at <http://ercoftac.mech.surrey.ac.uk>. Essentially these confirm that the thermocouple response time is adequate for the frequencies present in the cavity flow. Preliminary experiments with a fast response cold wire system by Cooper have confirmed similar levels of fluctuating temperatures.

A notable feature of the present results is the lack of any thermal stratification over the mid-height region of the cavity. The temperature on the vertical centre-line is effectively constant from $y/H = 0.3$ to 0.7 . Moreover, there is little variation in the mean and rms profiles taken across these levels. This suggests that the cavity is sufficiently tall for the mid-height region to approximate to that which one might expect in an infinitely tall cavity. If this is the case, then direct comparisons can be made with one-dimensional direct numerical simulations of the infinite cavity case (cf. Boudjemadi et al., 1997; Versteegh and Nieuwstadt, 1998, 1999). However care is needed in this assumption, as there could be residual effects from the top and bottom boundaries if the approach to full development is very shallow. These may account for the difference between the values of fluctuating temperatures reported here and those obtained by DNS in an infinitely high cavity. Confirmation would require the use of an even taller cavity.

4. Conclusions

The experiments described here have been designed to provide benchmark data against which various turbulence models can be tested. The provision of partially conducting top and bottom walls has successfully overcome the problems associated with trying to achieve an adiabatic surface there. The differing thermal stratification between the flows across the top and bottom, which has plagued previous work, has also been overcome, so that the experimental results are closely two-dimensional and anti-symmetric. Mean and turbulent temperature and velocity variations within the cavity have

Table 1
Summary of mid-height results

	Lower <i>Ra</i>	Higher <i>Ra</i>	
Cold wall temperature (°C)	15.1	15.6	
Hot wall temperature (°C)	34.7	54.7	
Av. wall temp. gradient (°C/m)	1540	3900	
Average Nusselt number	5.85	7.57	
Centre-line dT/dx (°C/m)	68	131	
Mid-cavity rms temperature (°C) (uncorrected)	0.92	1.48	
Max. vert. velocity (Av) (m/s)	0.139	0.191	
Mid-cavity rms, <i>v</i> (m/s)	0.10	0.134	
Mid-cavity d <i>V</i> /dx (s ⁻¹)	4.8	5.1	
Mid-cavity rms <i>u</i> (m/s)	0.053	0.077	
Mid-cavity <i>uv</i> (m ² /s ² × 10 ³)	2.4–2.8	4.2–5.0	
(<i>v_T</i> / <i>v</i>) _{<i>C/L</i>}	35	55	
(<i>α_T</i> / <i>α</i>) _{<i>C/L</i>}	23	30	
Variations of physical properties			
<i>Tabulated dry air values</i>			
Temperature (°C)	15.1	34.7	54.7
Thermal conductivity (W/mK × 10 ³)	25.3	26.8	28.3
β (K ⁻¹ × 10 ³)	3.47	3.25	3.05
ν (m ² /s × 10 ⁶)	14.6	16.5	18.4
Prandtl number	0.704	0.700	0.697
C _p (kJ/kg K)	—	1.005	—

been reported, together with heat fluxes and turbulent shear stresses. This data is also available to modellers through <http://ercoftac.mech.surrey.ac.uk>, where additional results are given, or more directly through <http://cfd.me.umist.ac.uk>.

For convenience, the results at the mid-height are summarized on Table 1. This table also contains derived results, such as average Nusselt number and an estimate of eddy viscosity at the centre-line, by comparing the Reynolds stress with viscous stress from the velocity gradient. A comparison of these provides an indication of the significant thickness of the viscous dominated region near the active walls. In the tall cavity, heat transfer is dominated by the turbulent interchange across the central region rather than by convection across the top and bottom.

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